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Inflammatory status in Holstein and Montbéliarde cows during the periparturient period and sequential feed restriction challenges

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ABSTRACT

The periparturient period of dairy cows is characterized by changes in innate inflammatory status (IS), but the physiological origin of these changes remains unclear. The objectives of this study were to investigate the effects of breed and feed efficiency on IS during the periparturient period, mid-lactation, and a sequential nutritional challenge (SNC) of 4 successive 4-d feed restriction (FR) periods. Fourteen Holstein (HOLS) and 14 Montbéliarde (MONT) multiparous cows were selected a posteriori from 40 animals based on extreme residual feed intake (RFI) during the first 10 wk of lactation and classified as RFI+ (0.42 to 2.34 kg DMI/d, n = 14) or RFI− (−2.38 to −0.44 kg DMI/d, n = 14). Cows were studied from wk −3 to +22 of lactation. Starting at 86 ± 9 DIM, cows underwent 4 periods of 4-d FR to meet 50% of individual net energy requirements (FR1, FR2, FR3, FR4) followed by 10, 3, 3, and 10 d recovery periods of ad libitum intake, respectively. Blood samples were collected from the coccygeal vessels before morning feeding on wk −3, +2, +12, and +22 of lactation and on d 0 and 4 of each FR. Plasma was analyzed using the Milliplex Bovine Cytokine/Chemokine Panel (Merck, Germany). For both breeds, tumor necrosis factor α (TNFα), IL-1α, IL-10, monocyte chemoattractant protein-1 (MCP-1) and IL-6 were significantly lower on wk +2 than wk −3 and remained steady through lactation, but IL-8 increased at wk +2 compared with wk −3. Concentrations in TNFα, IL-10, MCP-1 and IL-8 were higher in HOLS than MONT. Concentration in macrophage inflammatory protein −1β (MIP-1β) was higher in RFI+ than RFI−. The significant FR effects on innate IS were small and not systematic across FR1 to FR4. The SNC induced relatively small changes in cytokine and chemokine concentrations, indicating that negative energy balance did not affect IS, despite consistent metabolic responses. Breed, phenotypic feed efficiency and repeated feed restrictions had limited effects on IS

Key words: dairy cow, feed restriction, inflammation, residual feed intake

INTRODUCTION

In dairy cows, the periparturient period is characterized by shifts in hormonal and nutritional status with mobilization of body fat, proteins, and bone minerals to support the onset of lactation (Pires et al., 2013; Valldecabres et al., 2018). After parturition, dairy cows often experience negative energy balance despite a gradual increase in DMI (Butler et al., 1981). Calving itself can modulate the innate immune system (Hughes et al., 2014), and early lactation cows must balance the nutrient use between metabolic and inflammatory responses to sustain milk production. It remains unclear whether systemic inflammation contributes to or results from reduced feed intake, negative energy balance and lipomobilization during this period (Horst et al., 2021).

The effects of feed restriction (**FR**) on inflammatory responses have been investigated in dairy cows at different physiological stages around parturition (Pascottini et al., 2019; Esposito et al., 2020), but it remains unclear whether severe FR alone can induce systemic inflammation or whether it alters the cow's response to an immune or inflammatory challenge. To avoid the complex biological regulation associated with early lactation, mid-lactation models have been used to test the effects of FR on inflammatory responses to mammary challenges with LPS or *Streptococcus uberis*. Studies in mid-lactation cows reported limited effects of FR on animal-level responses to LPS or *S. uberis* challenges (Perkins et al., 2002; Moyes et al., 2009). Similarly, underfed ketotic Holstein cows challenged with LPS in early lactation (24 DIM) showed limited modifications of indicators of inflammatory responses compared with normally fed controls (Pires et al., 2019). Nonetheless, inflammatory responses to FR may depend on the severity of FR. In mid-lactation cows, concentrations of positive acute

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phase proteins such as haptoglobin, serum amyloid A, and LPS binding protein increased after 5 d of FR at 40% of energy requirements (**ER**), compared with 5 d at 60%, 80%, or 100% of ER (Kvidera et al., 2017b). In addition, correlations between markers of inflammation and gut barrier permeability have been reported (Horst et al., 2020), suggesting that increased intestinal permeability could contribute to systemic inflammation. In contrast, other studies did not detect inflammatory responses to experimental FR in late gestation or mid-lactation (Pascottini et al., 2019; Abeyta et al., 2023; Marins et al., 2023; Piantoni et al., 2023).

In the case of severe inflammation, the energy demand of activated immune cells may increase glucose consumption, leading to a decrease in milk production. Kvidera et al. (2017a) estimated that the amount of glucose required during the 12 h following an experimental LPS challenge to maintain euglycemia in mid-lactation cows exceeded 1 kg. A pro-inflammatory status may therefore divert limiting nutrients away from milk secretion, and affecting both production and efficiency of feed conversion into milk. It is therefore relevant to assess the relationships between innate inflammatory status (**IS**), breed and phenotypic feed efficiency, and whether breed or feed efficiency classification modulate an inflammatory response to FR.

This study used Holstein (**HOLS**) and Montbéliarde (**MONT**) cows from a previous experiment, where HOLS cows produced more ECM but had lower energy balance and greater plasma nonesterified fatty acids (**NEFA**), whereas MONT cows maintained greater BCS. These differences among species also appeared during the first phases of the sequential nutritional challenge (**SNC**) consisting of 4 periods of 4-d partial FR (Pires et al., 2025). The current study used a subset of HOLS and MONT cows divergent for phenotypic residual feed intake (**RFI**) within breeds. The objectives were to compare the innate IS of the cows during the periparturient period until mid-lactation, and during SNC performed near peak of lactation, to examine the differences between lactation stages, breeds and phenotypic RFI classification.

The hypotheses were (1) innate IS was modified during the periparturient period, (2) this modulation differed by breed and phenotypic feed efficiency classification (i.e., RFI), reflecting differing prioritization of nutrients for milk production versus nonproductive functions, (3) induced FR modulated the innate IS, with the responses changing across successive feed restrictions (FR1 to FR4), and (4) the SNC model allowed detection of differences in inflammatory robustness (i.e., magnitude and direction of response at each FR) between breeds and feed efficiency groups, reflecting differences in their ability to cope with successive FR. The IS was assessed by measuring a panel of cytokines and chemokines specific to innate immunity.

MATERIALS AND METHODS

Animals, Measurements, and Challenge

All procedures involving animals were approved by the Committee of the Auvergne-Rhône-Alpes region and the French Ministry of higher Education, Research and Innovation (APAFIS#16859-2015043014541577v7). The experiment was conducted in the Herbipôle Research Unit (INRAE, UE1414, Theix, France; http//doi.org/10.15454/1.5572318050509348E12).

Cows were part of a larger study involving 40 multiparous cows (18 HOLS and 22 MONT) conducted between September 2019 and May 2020 (Pires et al., 2025). Calvings occurred between October 3, 2019 and December 23, 2019, and cows were studied from 4 wk before expected calving to 22 wk of lactation (Figure 1). For this study, a subset of 28 cows comprising 14 HOLS and 14 MONT was selected based on extreme phenotypic RFI, classified as positive (RFI+, n = 14) or negative (RFI−, n = 14), with 7 cows selected per RFI group and breed. The RFI was calculated using data collected during the first 10 wk of lactation as the residual of the multiple regression:

$$\text{DMI kg/d} = \text{breed} + \text{milk energy secretion} + \text{BW}^{0.75} + \text{delta BW10wks},$$

where milk energy secretion (MJ/d) = milk yield (kg) × {0.44 + [0.0055 × (−40 + fat content; g/kg)]} × 7.12 (INRA, 2007); $\text{BW}^{0.75}$ = metabolic BW; delta BW10wks = BW at wk 10 of lactation minus BW at wk 1. The DMI, milk energy secretion, and $\text{BW}^{0.75}$ were averaged over the first 10 wk of lactation. Residual feed intake phenotyping resulted in DMI intervals of 0.42 to 2.34 kg/d for RFI+ and −2.38 to −0.44 kg/d for RFI−, and was completed before initiation of the SNC described below.

Holstein and Montbéliarde cows were studied from 4 wk before expecting calving to 22 wk of lactation. Cows were housed in the same freestall pen equipped with individual feed bunks and automatic gates (model Dairy Gates 3, Sodalec). They were offered TMR with forage to concentrate ratio of 88%:12% before calving, and 56%:44% after calving, with NEL content of 6.62 MJ/kg and 7.41 MJ/kg DM, respectively. Before and after calving, the amount of DM distributed was sufficient to allow 10% of refusals, ensuring ad libitum feed intake outside the FR periods. Feed was offered after morning milking and cows had free access to water. Pre- and postpartum TMR were collected weekly, pooled, and analyzed for nutrient composition (Pires et al., 2025). Cows were milked twice daily (starting at 0730 h and 1530 h) and milk yield was recorded daily. Cows were weighed once a week before calving and after each milking during lactation.

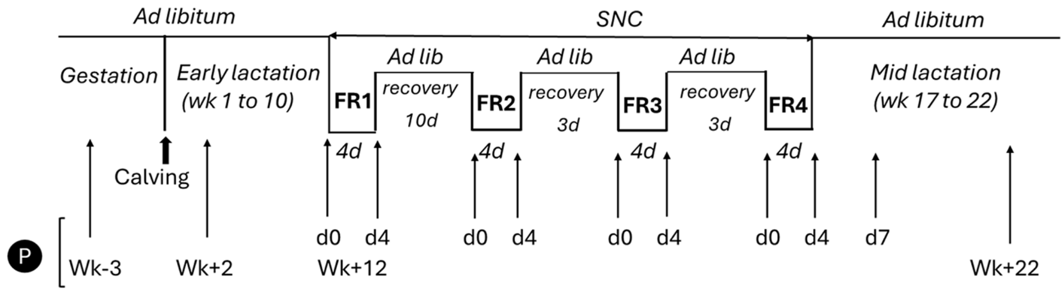

Figure 1. Schematic representation of the experimental design adapted from Pires et al. (2025). Multiparous Holstein and Montbéliarde cows were studied from 3 wk before calving (wk −3) to mid-lactation (wk +22). Before and after calving, the amount of DM distributed allowed 10% of refusals, ensuring ad libitum (Ad lib) feed. From the 12th wk (wk +12), cows underwent a sequential nutritional challenge (SNC) of 4 periods of 4-d feed restriction (FR1, FR2, FR3, FR4) during which feed allowance was restricted to meet 50% of individual NEL requirements (see Pires et al., 2025). Plasma samples were produced at (mean ± SD) −16 ± 5 DIM (wk −3), at 13 ± 2 DIM (wk +2), at 86 ± 9 DIM (wk +12), and at 155 ± 10 DIM (wk +22). Plasma was also sampled during (SNC) before and after feed restriction (FR) at d 0 and d 4 of FR1, FR2, FR3, and FR4, and 3 d after the FR of FR4 (d 7 of FR4).

During RFI phenotyping, milk samples were collected at 4 consecutive milkings each week and analyzed for milk composition. Offered feed and refusals were weighed and subsamples were collected 5 d/wk to determine DM content and calculate DMI (Pires et al., 2025).

An experimental SNC composed of a sequence of 4 FR periods (FR1, FR2, FR3, FR4), which consisted of 4 d of 50% reduction in energy intake allowance (d 0 to 4) followed by recovery periods of 10, 3, 3, and 10 d of ad libitum intake to exceed ER, respectively, was initiated at 86 ± 9 DIM (near peak lactation) for the first FR period (Figure 1). During SNC periods, feed offered and refused were weighed daily and milk composition measured daily. Each animal was its own control and energy restriction was based on individual needs according to BW and milk production and composition (registered the wk before the first FR (Pires et al., 2025). Body condition score was recorded once a week and the d 0 and 4 of restriction periods. Cows returned to ad libitum intake after FR4. For more details on animal management and feeding, see Pires et al. (2025).

Blood Sampling and Analyses

Blood samples for plasma cytokine and chemokine analyses were collected from coccygeal vessels in EDTA tubes (1.95 mg/mL; Terumo Europe NV, Leuven, Belgium) in the morning before feeding, 3 wk before calving (wk −3; −16 ± 5 DIM), in early lactation after morning milking and before feeding (wk +2; 13 ± 2 DIM), near the lactation peak (wk +12; 86 ± 9 DIM), and in mid-lactation (wk +22; 155 ± 10 DIM, mean ± SD). During the FR periods (FR1, FR2, FR3, FR4), blood samples were collected before (d 0) and at the end of FR (d 4) for FR1, FR2, FR3, and FR4, and 3 d after the end of FR4 (d 7). As shown in Figure 1 illustrating the experimental design, sampling at d0 of FR1 corresponded to the same sample as wk +12. Within the 30 min following blood collection, plasma was obtained by centrifugation (1,400 × *g*, 15 min, 4°C), and conserved at −80°C before analyses. The lipomobilization was assessed on the plasma samples collected for cytokine and chemokine analyses by nonesterified fatty acid analysis (NEFA, Fujifilm Wako Chemicals Europe GmbH, Neuss, Germany). Plasma antioxidant capacity was assessed by Ferric reducing ability of plasma (**FRAP**) measurement (expressed in m*M* of Fe2+ equivalent/L; Dunière et al., 2021). Ferric reducing ability of plasma was measured at wk −3, wk +2, wk +12 (d 0 and d 4 of FR1). Intra- and interassay coefficients of variation were 3.0% and 8.4% for FRAP, and 4.0% and 4.6% for NEFA, respectively.

Plasma cytokine/chemokine analysis and quantification: Pro-inflammatory cytokines IL-1β, IL-1α, IL-17α, tumor necrosis factor α (TNFα), pro- and anti-inflammatory IL-6, and anti-inflammatory IL-10; pro-inflammatory chemokines IL-8 (C-X-C pattern), monocyte chemoattractant protein-1 or CCL2 (MCP-1), macrophage inflammatory protein −1β or CCL4 (MIP-1β), and vascular endothelial growth factor-A (VEGF-A) were assayed using Milliplex Bovine Cytokine/Chemokine Magnetic Bead Panel 1 (BCYT1–33K-10, Merck, Darmstadt, Germany) according to the manufacturer's instructions. Before the assays, individual plasma samples and homemade controls were diluted 1:2 in assay buffer. The plates were washed using an automatic wash station (Bio-Plex ProTM II Wash Station, Bio-Rad Laboratories, Hercules, CA), and were analyzed on a Luminex 200TM (Bio-Rad Laboratories, Hercules, CA). Finally, data were processed using the Bio-Plex Manager software 6.0 (Bio-Rad Laboratories, Hercules, CA). For each of the 10 cytokines and chemokines, the software calculated a Logistic-5PL type regression from the fluorescence intensity measured for the 7 standards, and returned the parameters of each regression (equation of the standard curve, R^2, low and high limits of quantification) that were checked for the range of recovery of the different concentrations. The results of sample concentrations were expressed in picograms per milliliter and values obtained between the detection and quantification limits were specified.

Performances of the Milliplex assay: Homemade plasma controls were obtained from dairy HOLS cows at 3 physiological stages. Blood samples of 3 animals were drawn the same day and pooled to obtain plasma G (= gestation, at 38 (SD = 5) d before calving). After calving, blood samples of 2 cows among the 3 previously used to obtain plasma G (the third cow having calved too early) were drawn the same day and pooled to obtain plasma L (= lactation, at 3 (SD = 1) d of lactation) and plasma PL (= peak of lactation, at 59 (SD = 1) d of lactation). Each plasma control was analyzed 20 times across four 96-well plates. The results of homemade plasma controls variability are presented in Supplemental Table S1 (see Notes). IL-1β was only detectable and quantifiable in PL in only 18 (including 4 concentrations below the low limit of quantification [**LLOQ**]) of the 20 determinations with high intra-assay variability (24%) and intermediate fidelity (86%). Only 10% of IL-6 concentrations in PL were comprised between the limit of detection (**LOD**) and LLOQ, the others being undetectable. For IL-17α, 40% and 55% of concentrations in G and L, respectively, were comprised between the LOD and LLOQ, the others being undetectable. For the other parameters, the means ± SD LOQ were 0.31 ± 0.03 pg/mL (IL-1α), 22.1 ± 26 pg/mL (IL-6), 45.5 ± 21 pg/mL (IL-8), 3.67 ± 1.93 pg/mL (IL-10), 30.7 ± 22.1 pg/mL (MCP-1), 1.95 ± 0.11 pg/mL (MIP-1β), 47.7 ± 25 pg/mL (TNFα), and 1.20 ± 0.18 pg/mL (VEGF-A). The ranges of values for intra-assay variation and intermediate fidelity were 1.76% to 7.73% and 3.82% to 12.9%, respectively.

Statistical Analysis

Statistical analyses were performed using SAS (version 9.4; SAS Institute Inc.) and XLSTAT Basic 2024.2.1 for nonparametric analyses. Natural logarithmic transformation of continuous variables was used when needed to comply with the assumptions of normality and homoscedasticity, which were assessed by data normality tests, residual analyses using UNIVARIATE procedure, and residual diagnostics options of MIXED procedure. For variables that did not meet normal distribution even after transformation (i.e., energy balance [**EB**], NEFA, FRAP, IL-1α, IL-6), a nonparametric Friedman test was applied to test the effect of lactation week (Hollander and Wolfe, 1973), and this is specified in Tables 1 and 2.

For all the variables (DMI, ECM, EB, BW, NEFA, FRAP, TNFα, IL-1α, IL-10, VEGF-A, MCP-1, IL-8, IL-6, and MIP-1β) throughout periparturient and mid-lactation periods: the repeated statement was used in the MIXED procedure to analyze the fixed effects of wk, breed, RFI and their interactions. Because breed and RFI may not be completely independent, breed and RFI effects and their interaction with week were analyzed in 2 separated models. Cow was included as the random effect, with repeated measures modeled using a spatial power [SP(POW)] covariance structure. Denominator degrees of freedom were estimated with the Kenward–Roger method. For the analysis of cytokines and chemokines, several ELISA plates were used to determine the plasma concentrations of all the samples. The plate effect was initially included in the MIXED procedures of SAS, but was subsequently removed due to the absence of significant effect. The associations between cytokines/chemokines and production parameters (DMI, ECM, EB, BW) and FRAP and NEFA were explored by Spearman correlation.

Concerning the data collected throughout FR2, FR3, and FR4 of the SNC, plasma cytokine and chemokine analyses were performed 18 mo after the analyses performed on samples of FR1 period. During this time, ELISA kits from different production lots were used. Consequently, the effects of dietary restriction were analyzed separately within each FR period (FR1, FR2, FR3, and FR4), and comparisons between FR periods were not performed. All the variables (DMI, ECM, EB, BW, NEFA, FRAP, TNFα, IL-1α, IL-10, VEGF-A, MCP-1, IL-8, IL-6, and MIP-1β) were analyzed using the MIXED procedure in SAS with the fixed effects of day (d 4 = feed restriction vs. d 0 = control), breed, RFI and their interactions, in 2 separate models for the aforementioned reasons. Cow was included as the random effect. The differences between d 4 and d 0 of FR1, FR2, FR3, and FR4 were calculated for all the variables, and paired *t*-tests used to test the null hypothesis (i.e., differences between the means are equal to zero, with $\alpha = 0.05$).

The data are reported as LSM with 95% CI, and transformed data were back-transformed to the original scale and presented as geometric means with 95% CI. Statistical differences were considered significant at $P \leq 0.05$, and trends of significance for $0.05 < P \leq 0.1$.

RESULTS

Periparturient and Mid-Lactation Periods, Effects of Breed, and RFI

Animal Performances. Daily DMI increased ($P < 0.001$) from 14.5 kg/d (95% CI = 13.4–15.7 kg/d) before calving (wk −3) to 21.8 kg/d (95% CI = 20.6–22.9

Table 1. Effects of lactation wk and breed on production variables measured in multiparous Holstein (HOLS; n = 14) and Montbéliarde (MONT; n = 14) at wk −3, 2, 12, and 22 of lactation

	Lactation week				*P*-value[1]		
Item[2]	wk −3	wk +2	wk +12	wk +22	Week	Breed	Week × breed
DMI, kg/d							
HOLS	15.2	22.4	26.1	24.3	<0.001	0.005	0.26
Lower CI	13.5	20.8	24.5	22.5			
Upper CI	16.9	24.0	27.8	26.2			
MONT	13.9	21.1	23.0	21.9			
Lower CI	12.2	19.5	21.4	19.8			
Upper CI	15.5	22.7	24.6	24.0			
Week effect[3]	C	B	A	AB			
ECM, kg/d							
HOLS	—	37.8	37.5	32.8	0.006	0.005	0.59
Lower CI		35.2	34.8	29.7			
Upper CI		40.5	40.1	35.9			
MONT	—	32.3	33.4	29.6			
Lower CI		29.6	30.7	26.2			
Upper CI		34.9	36.0	33.0			
Week effect[3]		A	A	B			
EB, MJ/d							
HOLS	32.7	−8.8	16.6	19.8	<0.001	0.34	0.31
Lower CI	22.3	−18.8	6.6	7.9			
Upper CI	43.2	1.3	26.7	31.6			
MONT	25.6	0.7	9.0	11.6			
Lower CI	15.6	−9.4	−1.0	−1.7			
Upper CI	35.6	10.7	19.1	24.9			
Week effect[4]	A	C	B	B			

[1]The repeated statement of the MIXED procedure was used to analyze the fixed effects of week, breed, and their interactions. Cow was included as the random effect, with repeated measures modeled using a spatial power [SP(POW)] covariance structure. Denominator degrees of freedom were estimated with the Kenward–Roger method.

[2]Values are LSM with 95% CI.

[3]Main week effects not sharing a common letter (A–C) within a row differ ($P \leq 0.05$). The bottom row for each variable refers to statistical comparisons between week across both breeds.

[4]The variable never met a normal distribution, even after log-transformation. A nonparametric Friedman test was applied to confirm the effect of lactation week.

kg/d) in wk +2, increased ($P < 0.001$) from wk +2 to 24.6 kg/d (95% CI = 23.4–25.7 kg/d) at wk +12, and then tended to decrease ($P = 0.084$) to 23.1 kg/d (95% CI = 21.7–24.5 kg/d) in mid-lactation (wk +22) compared with wk +12. Dry matter intake was higher in HOLS than in MONT (22.0 kg/d [95% CI = 21.0–23.0] vs. 20.0 kg/d [95% CI = 19.0–21.0], respectively, $P = 0.005$, Table 1), and per definition, DMI was higher for RFI+ than RFI− cows (21.6 kg/d [95% CI = 20.7–22.5] vs. 19.3 kg/d [95% CI = 18.4–20.1], respectively, $P < 0.001$, Supplemental Table S2, see Notes). The RFI groups averaged +1.24 (SD = 0.17) kg DMI /d (RFI+; n = 14, with 7 HOLS and 7 MONT) and −1.13 kg DMI/d (SD = 0.13; RFI−; n = 14, with 7 HOLS and 7 MONT). Energy-corrected milk was higher in HOLS than in MONT (36.0 kg/d [95% CI = 34.0–38.0] vs. 31.7 kg/d [95% CI = 29.7–33.8], respectively, $P = 0.005$, Table 1), and EB, which decreased sharply in early lactation (wk +2), did not differ between HOLS and MONT, but was higher in RFI+ than in RFI− cows (19.7 MJ/d [95% CI = 15.1–24.4] vs. 6.81 MJ/d [95% CI = 1.70–11.9], respectively, $P < 0.001$, Supplemental Table S2). Holsteins and MONT had similar BW ($P = 0.36$, not shown) both 3 wk before calving (770 kg [95% CI = 741–799] and 737 kg [95% CI = 708–765], for HOLS and MONT, respectively) and during lactation (ranging from 684 kg [95% CI = 655–713] to 706 kg [95% CI = 676–737] in HOLS, and from 661 kg [95% CI = 632–690] to 704 kg [95% CI = 674–735] in MONT between wk +2 and wk +22, respectively).

Metabolic Status. Plasma NEFA concentrations were increased significantly in early lactation (wk +2) by 310% versus wk −3 (0.095 m*M* [95% CI = 0.053–0.136], $P < 0.001$), by 231% versus wk +12 (0.117 m*M* [95% CI = 0.076–0.158], $P < 0.001$), and by 392% versus wk +22 (0.079 m*M* [95% CI = 0.038–0.120], $P < 0.001$) (Table 2). Ferric reducing ability of plasma was increased significantly near the lactation peak (wk +12) by 6.7% versus wk −3 (0.223 m*M* [95% CI = 0.214–0.232], $P = 0.005$) and by 10.4% versus wk +2 (0.216 m*M* [95% CI = 0.207–0.225], $P < 0.001$), but did not differ between wk −3 and wk +2 (Table 2). For both NEFA and FRAP,

Table 2. Effects of lactation week and breed on plasma NEFA, FRAP, cytokines, and chemokines measured in multiparous Holstein (HOLS; n = 14) and Montbéliarde (MONT; n = 14) before calving, in early lactation, near the lactation peak, and in mid-lactation

	Lactation wk				*P*-value[1]		
Item[2]	wk −3	wk +2	wk +12	wk +22	Week	Breed	Week × breed
NEFA,[3] m*M*							
HOLS	0.097	0.426	0.119	0.062	<0.001	0.57	0.31
Lower CI	0.038	0.368	0.061	0.003			
Upper CI	0.155	0.484	0.177	0.120			
MONT	0.093	0.351	0.115	0.096			
Lower CI	0.035	0.293	0.057	0.038			
Upper CI	0.151	0.409	0.174	0.155			
Week effect[4]	B	A	B	B			
FRAP,[3] m*M*							
HOLS	0.212	0.215	0.234	0.235	<0.001	0.33	0.072
Lower CI	0.199	0.202	0.221	0.222			
Upper CI	0.225	0.228	0.247	0.248			
MONT	0.234	0.216	0.242	0.230			
Lower CI	0.221	0.203	0.229	0.218			
Upper CI	0.247	0.229	0.255	0.243			
Week effect[4]	BC	C	A	AB			
TNFα,[5] pg/mL							
HOLS	6,626	5,385	4,637	3,782	0.0019	0.034*	0.21
Lower CI	3,634	2,954	2,543	2,074			
Upper CI	12,081	9,818	8,455	6,895			
MONT	2,921	1,634	2,514	2,408			
Lower CI	1,602	896	1,379	1,321			
Upper CI	5,325	2,979	4,583	4,390			
Week effect[4]	A	B	AB	AB			
IL-1α,[3] pg/mL							
HOLS	60.2	54.0	41.9	36.6	<0.001	0.19	0.40
Lower CI	41.2	35.0	22.9	17.6			
Upper CI	79.2	73.0	60.9	55.7			
MONT	47.5	26.7	30.6	26.6			
Lower CI	28.5	7.7	11.6	7.6			
Upper CI	66.6	45.7	49.6	45.6			
Week effect[4]	A	B	B	B			
IL-10,[5] pg/mL							
HOLS	390	327	257	202	<0.001	0.031*	0.13
Lower CI	253	212	167	131			
Upper CI	602	505	397	312			
MONT	183	137	159	136			
Lower CI	119	89	103	88			
Upper CI	283	211	246	210			
Week effect[4]	A	BC	B	C			
VEGF-A,[5] pg/mL							
HOLS	89.8	98.8	72.9	66.9	0.001	0.09	0.09
Lower CI	60.6	66.6	49.1	45.1			
Upper CI	133.3	146.6	108.2	99.3			
MONT	58.4	50.6	48.3	49.2			
Lower CI	39.4	34.1	32.6	33.2			
Upper CI	86.6	75.1	71.7	73.0			
Week effect[4]	A	A	B	B			
MCP-1,[5] pg/mL							
HOLS	990	853	801	735	0.002	0.022*	0.87
Lower CI	752	647	608	558			
Upper CI	1,305	1,123	1,056	968			
MONT	627	528	552	502			
Lower CI	476	401	419	381			
Upper CI	826	696	727	661			
Week effect[4]	A	B	B	B			
IL-8,[5] pg/mL							
HOLS	976	1,596	1,084	941	<0.001	0.015*	0.45
Lower CI	680	1,112	755	656			
Upper CI	1401	2,291	1,555	1,350			
MONT	513	1,000	554	696			
Lower CI	358	697	386	485			
Upper CI	736	1,435	795	998			

Continued

Table 2 (Continued). Effects of lactation week and breed on plasma NEFA, FRAP, cytokines, and chemokines measured in multiparous Holstein (HOLS; n = 14) and Montbéliarde (MONT; n = 14) before calving, in early lactation, near the lactation peak, and in mid-lactation

	Lactation wk				*P*-value[1]		
Item[2]	wk −3	wk +2	wk +12	wk +22	Week	Breed	Week × breed
Week effect[4]	B	A	B	B			
IL-6,[3] pg/mL							
HOLS	786	270	20.9	20.1	<0.001	0.68	0.34
Lower CI	562	47	−202.8	−203.6			
Upper CI	1,010	494	244.6	243.8			
MONT	1,060	167	2.7	9.1			
Lower CI	836	−57	−221.0	−214.6			
Upper CI	1,283	390	226.4	232.8			
Week effect[4]	A	B	C	C			
MIP-1β,[5] pg/mL							
HOLS	106	100	88	87	0.10	0.85	0.42
Lower CI	81	76	67	66			
Upper CI	139	131	115	114			
MONT	110	89	97	96			
Lower CI	84	68	74	73			
Upper CI	144	117	127	126			

[1]The repeated statement of the MIXED procedure was used to analyze the fixed effects of week, breed, and their interactions. Cow was included as the random effect, with repeated measures modeled using a spatial power [SP(POW)] covariance structure. Denominator degrees of freedom were estimated with the Kenward–Roger method.

[2]FRAP = ferric reducing ability of plasma, MCP-1 = monocyte chemoattractant protein-1 or CCL2, MIP-1β = macrophage inflammatory protein −1β or CCL4, NEFA = nonesterified fatty acids, TNFα = tumor necrosis factor α, VEGF-A = vascular endothelial growth factor-A.

[3]For NEFA, FRAP, IL-1α, and IL-6, values are LSM with 95% CI. The variable never met a normal distribution, even after log-transformation. A nonparametric Friedman test was applied to confirm the effect of lactation week.

[4]Main week effects not sharing a common letter (A–C) within a row differ ($P \leq 0.05$). The bottom row for each variable refers to statistical comparisons between week across both breeds.

[5]For TNFα, IL-10, VEGF-A, MCP-1, IL-8, and MIP-1β, values are back-transformed and presented as geometric means with 95% CI.

*Significant differences between breeds with HOLS > MONT ($P \leq 0.05$).

no significant differences were observed between HOLS and MONT, or RFI+ and RFI− cows (Table 2; Supplemental Table S3, see Notes).

Breed and RFI Differences of Plasma Concentrations of Cytokines and Chemokines. These results are presented in Figure 2 and Table 2. During the periparturient period (wk −3 and wk +2), TNFα, IL-1α, IL-10, MCP-1, and IL-6 decreased significantly (−32.6% [$P = 0.002$], −25% [$P = 0.038$], −21% [$P < 0.001$], −15% [$P = 0.002$], and −76% [$P < 0.001$], respectively). During lactation, TNFα, IL-1α, and MCP-1 remained stable over time, IL-10 decreased gradually between wk +2, wk +12 and wk +22 (−22% between wk +2 and wk +22, $P = 0.051$). IL-6 decreased sharply between wk +2 and wk +12 (−94.6%, $P < 0.001$) and remained low at wk +22. However, out of 28 samples, only 9 and 12 were measurable for IL-6 at wk +12 and wk +22, respectively. In contrast to the other cytokines and chemokines, IL-8 increased during the periparturient period between wk −3 and wk +2 (+78.5%, $P < 0.001$) and decreased at wk +12 (−39%, $P < 0.001$) and wk +22 (−36%, $P < 0.001$) compared with wk +2, to return to the levels observed before calving. There was no significant change in MIP-1β over time, but it was higher for RFI+ than for RFI− cows (+42.6%, $P = 0.024$, Supplemental Table S3, Figure 2). VEGF-A did not change significantly between wk −3 and wk +2, but decreased at wk +12 compared with wk −3 and wk +2 (−18%, $P = 0.007$ and −16%, $P = 0.017$, respectively) and remained stable between wk +12 and wk +22. Breed differences were limited to higher concentrations in TNFα (+115.7%, $P = 0.034$), IL-10 (+87.0%, $P = 0.031$), MCP-1 (+52.6%, $P = 0.022$), and IL-8 (+68.4%, $P = 0.015$) for HOLS than for MONT, with no significant breed by week interactions (Table 2). IL-1β and IL-17α were excluded from the data analyses because the majority of values (82% and 62%, respectively) were undetectable or below the limit of quantitation (Supplemental Table S1).

Effects of FR Periods by Breed and RFI Group

Animal Performance and Metabolic Status. By design, FR caused significant ($P < 0.001$) and repeatable decreases in ECM, DMI, and EB (Figure 3, Supplemental Table S4, see Notes). On average, across all 4 FR and on d 4 of FR compared with pre-restriction d 0 (d 4 vs. d 0), ECM decreased by 6.2 kg/d (SD = 0.4; −18.2%, SD = 1.2), DMI by 10.8 kg/d (SD = 0.4; −45.8%, SD = 1.0), and EB by 53.5 MJ/d (SD = 2.5). Plasma NEFA increased consistently (+0.436 m*M* [SD =

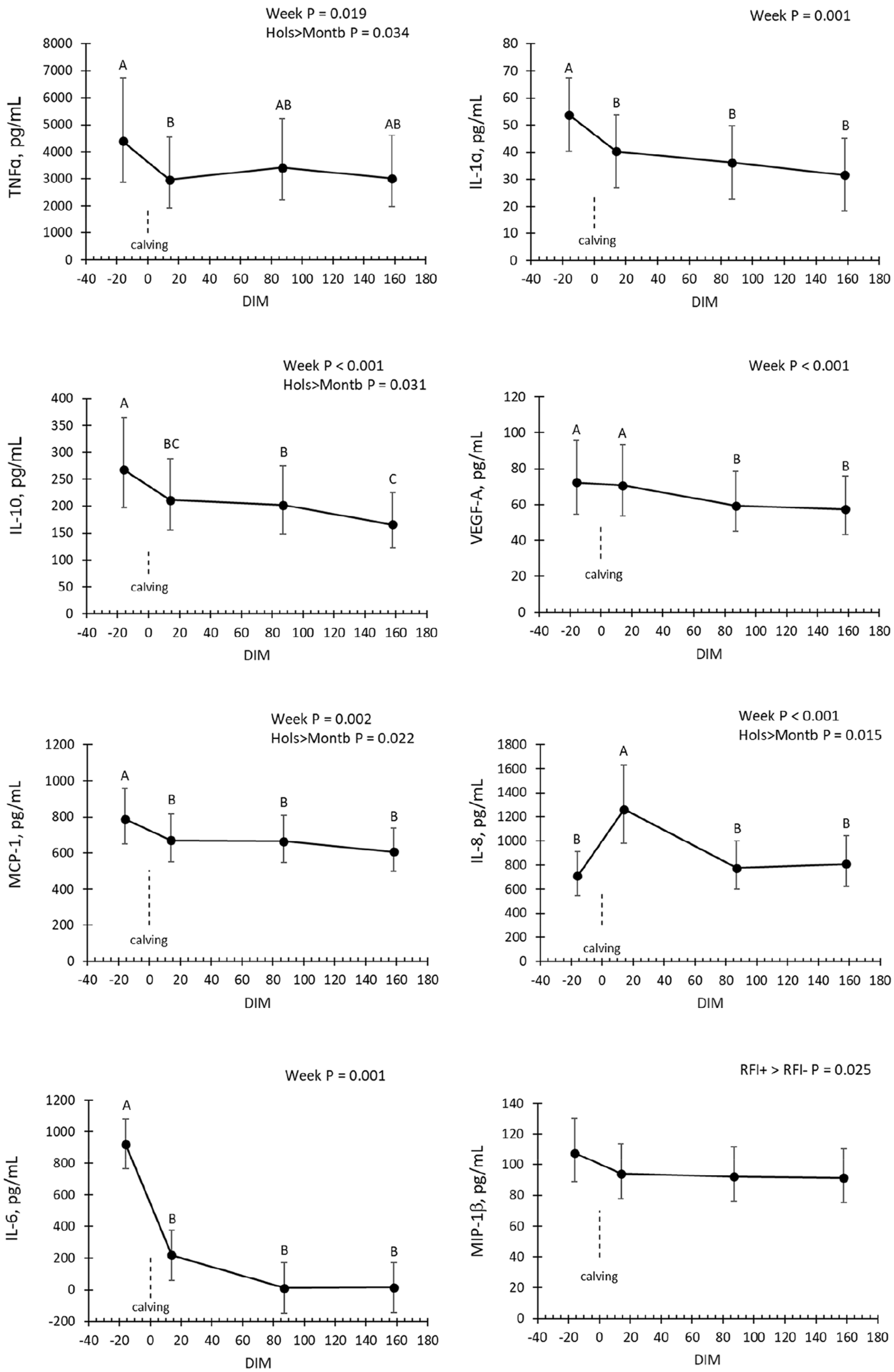

Figure 2. Cytokines and chemokines tumor necrosis factor α (TNFα), IL-1α, IL-10, vascular endothelial growth factor-A (VEGF-A), monocyte chemoattractant protein-1 or CCL2 (MCP-1), chemokine C-X-C pattern (IL-8), IL-6, and macrophage inflammatory protein −1β or CCL4 (MIP-1β), in multiparous Holstein (HOLS; n = 14) and Montbéliarde (MONT; n = 14) either residual feed intake (RFI) plus (RFI+; 0.42 to 2.34 kg DMI/d) or RFI− (−2.38 to −0.44 kg DMI/d) at (mean ± SD) wk −3 (−16 ± 5 DIM), wk +2 (13 ± 2 DIM), wk +12 (86 ± 9 DIM), and wk +22 (155 ± 28 DIM) relative to calving. Different letters (A–C) indicate significant differences between the means, $P \leq 0.05$. Variables TNFα, IL-10, VEFGF-A, MCP-1, IL-8, and MIP-1β met normality after log-transformation, and the reported values are back-transformed and presented as geometric means with 95% CI. For IL-1α and IL-6 that never met a normal distribution, even after log-transformation, values are LSM with 95% CI.

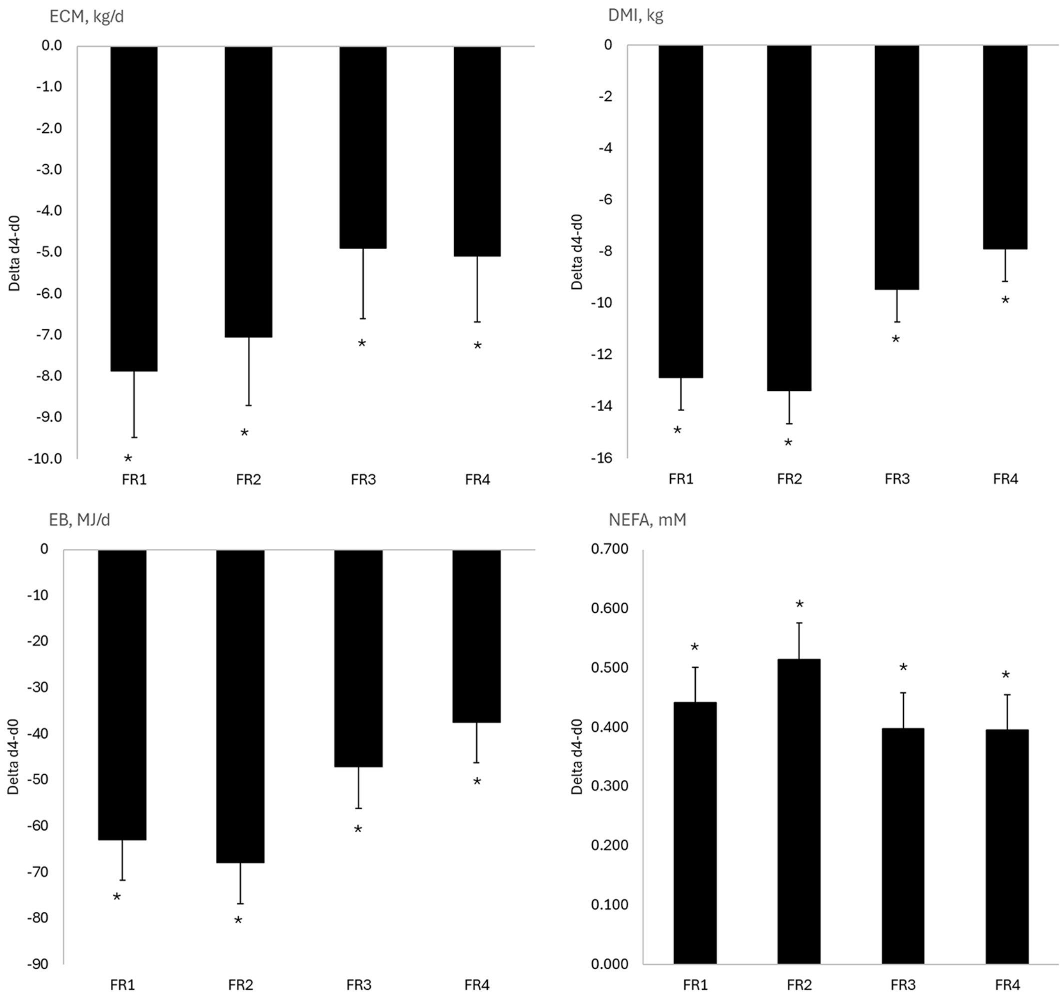

Figure 3. LSM with 95% CI of ECM, DMI, energy balance (EB), and nonesterified fatty acids (NEFA) in multiparous Holstein (HOLS; n = 14) and Montbéliarde (MONT; n = 14), between the wk 12 and 17 of lactation, subjected to a sequential nutritional challenge of 4 successive 4-d 50% energy requirement periods (FR1, FR2, FR3, FR4), where FR = feed restriction. * Indicates that differences between d 4 and d 0 were significant ($P \leq 0.05$).

0.016]; +623.8% [SD = 24.0], $P < 0.001$). Ferric reducing ability of plasma was measured only during FR1 and decreased by 12.7% (SD = 2.3; $P < 0.001$).

Innate Inflammatory Status. As shown in Figure 4 and Table 3, FR1 significantly decreased the proinflammatory chemokine MCP-1 (−9.3%, $P = 0.001$) and the anti-inflammatory cytokine IL-10 (−7.3%, $P < 0.05$). During FR1, IL-1α decreased in HOLS (−7.9%, $P = 0.007$) but not in MONT (day × breed $P = 0.01$, Table 3 and Supplemental Table S5, see Notes), and MIP-1β decreased in RFI+ (−19.3%, $P < 0.001$) but not in RFI− (day × RFI, $P = 0.003$, Supplemental Table S6, see Notes). During FR2, only IL-10 decreased (−9.4%, $P < 0.01$). FR3 increased TNFα (+8.5%, $P = 0.01$), IL-1α (+7.7%, $P = 0.06$), VEGF-A (+8.5%, $P < 0.05$), and IL-8 (+48.4%, $P = 0.001$), respectively (Figure 4, Table 3). During FR4, IL-8 increased (+38%, $P = 0.01$), whereas IL-10 and MCP-1 decreased (−14.9% and −9.9%, respectively both $P < 0.001$). IL-8 concentrations during FR1 were higher in HOLS than in MONT (+64.7%, $P = 0.02$). MIP-1β concentrations were higher in RFI+ than in RFI− cows during FR2 (+44.7%, $P = 0.03$), with tendencies for an RFI effect in FR3 (+58.4%, $P = 0.09$) and FR4 (+39.2%, $P = 0.07$; Supplemental Table S6).

Analysis of Correlations. As presented in Supplemental Figure S1 (see Notes), IL-1α, IL-10, MCP-1, TNFα and VEGF-A were highly correlated ($0.64 < r < 0.91$, $P < 0.001$) across all sampling time points (i.e., wk −3, wk +2, d 0 FR1, d 4 FR1, d 0 FR2, d 4 FR2, d 0 FR3, d 4 FR3, d 0 FR4, d 4 FR4 and wk +22). Considering the sampling time points wk −3, wk +2, d 0 FR1, d 4 FR1, and wk +22, FRAP was positively correlated with EB (0.28, $P < 0.001$) and ECM (0.37, $P < 0.001$), and negatively correlated with NEFA (−0.41, $P < 0.001$).

DISCUSSION

We selected 28 HOLS and MONT cows with extreme phenotypic feed efficiency (RFI+ or RFI−) from a previous experiment (Pires et al., 2025) to study the innate IS and its regulation across lactation and during 4 consecutive 4-d FR. Inflammatory status was assessed using a panel of plasma cytokines and chemokines. Cytokine profiles changed over time during the periparturient period, with most cytokines decreasing at wk +2 compared with prepartum, except IL-8, which increased. Although overall concentrations of TNFα, IL-10, MCP-1, and IL-8 were higher in HOLS than in MONT, temporal cytokine patterns were not different across breeds, and were not affected by phenotypic feed efficiency (except for higher MIP-1β in RFI+). Repeated short FR did not alter IS, in contrast with the consistent negative EB and metabolic effects of FR. Therefore, in accordance with hypothesis 1, the innate IS was modified during the periparturient period, and in contrast to the hypothesis 2, this temporal modulation did not differ according to breed or RFI. The hypothesis 3 was weakly supported with small FR effects. In contrast to hypothesis 4, the few significant interactions between breed or RFI and FR were nonreproducible throughout the SNC and did not allow us to demonstrate possible differences in inflammatory robustness among breeds or RFI.

Innate IS from Peripartum to Mid-Lactation According to Breed and RFI

The decreases in TNFα, IL-1α, IL-10, IL-6, and MCP-1 observed between 3 wk before and 2 wk after calving in HOLS and MONT cows agree with previous studies reporting a decrease in IL-1β, IL-6, and TNFα in early lactation with a nadir between 7 and 14 d relative to calving (Trevisi et al., 2015; Lopreiato et al., 2019; Kuhla, 2020). During the early stages of lactation, the inflammatory status of cows can be triggered by noninfectious factors such as placental expulsion, uterine involution, and adipose tissue remodeling (Trevisi et al., 2025), as well as by the uterus's defense against bacterial infections following a normal delivery (Singh et al., 2008). Conversely, inflammatory stimulation may stimulate cortisol secretion, which can have an immunosuppressive effect and explain the subsequent decrease in inflammatory markers such as cytokines and chemokines a few weeks after calving (Hughes et al., 2014; Trevisi et al., 2015). In the present study, IL-6 showed the most pronounced decrease between the wk −3 and the wk +2, reaching baseline values at wk +12. The high IL-6 concentration in late gestation and its subsequent decrease in early lactation may be explained by high IL-6 concentrations in the bovine uterus before calving, which return to baseline by 8 d after calving (Ishikawa et al., 2004). High IL-6 concentration may also contribute to hyper-responsiveness of the hypothalamic-pituitary-adrenal axis and stimulate cortisol synthesis (Kabaroff et al., 2006). The concomitant decrease in the anti-inflammatory cytokine IL-10 and the pro-inflammatory cytokine TNFα during the periparturient period further suggests the onset of a pro-inflammatory state during early lactation (Lee et al., 2024; Trevisi et al., 2025). Pro-inflammatory cytokines can induce liver synthesis of amyloid A (Ceciliani et al., 2012), a positive acute phase protein with cytokine-like properties that stimulates IL-8 transcription via NF-κB activation (He et al., 2003). This mechanism may regulate the temporal dynamics of plasma cytokines, as recently observed across lactation stages in dairy cows (Tarar et al., 2025). It may also explain the distinct temporal profile of IL-8 observed in the present study, which contrasted with the synchronous decrease of other cytokines and chemokines from wk −3 to wk +2, and with the significant correlations observed among IL-1α, IL-10, MCP-1, TNFα, and VEGF-A, but not with IL-8. Recent studies have shown that the immune dysfunction in early lactating cows does not indicate a reduced ability to fight infection. On the contrary, inflammation is highly inducible in early lactating cows (Cortese et al., 2024; Opgenorth et al., 2024b).

In the present study, the plasma antioxidant capacity was reduced at wk +2 compared with wk +12 and wk +22, as indicated by FRAP. This reduction in antioxidant capacity in early lactation may favor the persistence of an IS or, in combination increased circulating NEFA, may impair leukocyte function through excessive reactive oxygen species production during mitochondrial β-oxidation (Contreras and Sordillo, 2011). Furthermore, in the present study we observed marked interindividual variation in cytokine profiles, in agreement with previous research (Trevisi et al., 2025), likely reflecting differences in the ability of cows to maintain homeostasis during the periparturient period, when nutrient demands for milk production and immune response may compete. Indeed, although IL-6 decreased at wk +2, 2 HOLS and 2 MONT maintained elevated TNFα and MCP-1 at wk +2, suggesting likely inflammation despite absence of clinical events. In 3 of the 4 cows, inflammation resolved by wk +12, whereas 1 cow maintained elevated concentrations of TNFα, MCP-1, IL-1α, VEGF-A, and IL-10 through wk +22.

Breed and RFI Effects on IS

Holstein and MONT were chosen for their known differences in dairy potential and nutrient prioritization. As expected, HOLS had higher milk yield and DMI

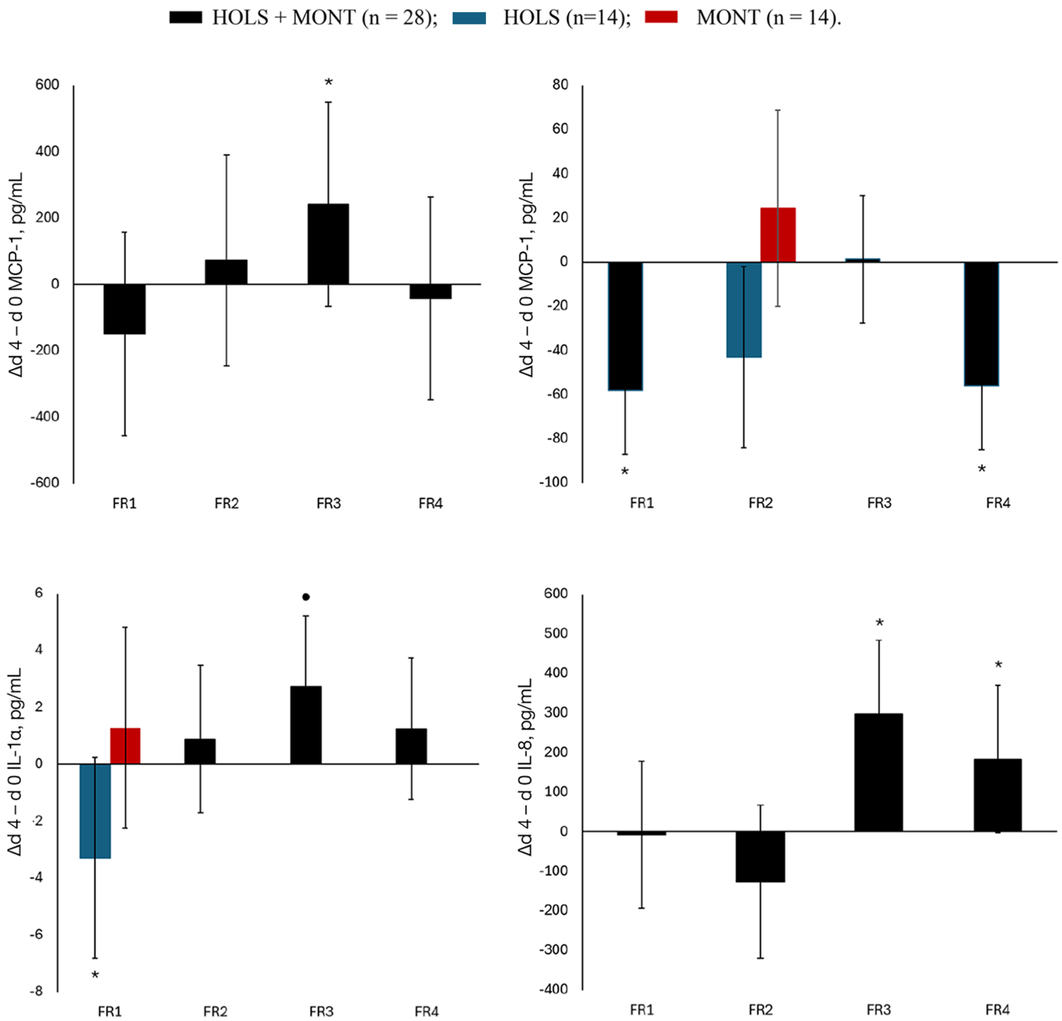

Figure 4. LSM with 95% CI of cytokines and chemokines tumor necrosis factor α (TNFα), IL-1α, IL-10, vascular endothelial growth factor-A (VEGF-A), monocyte chemoattractant protein-1 or CCL2 (MCP-1), chemokine C-X-C pattern (IL-8), IL-6, and macrophage inflammatory protein −1β or CCL4 (MIP-1β), in multiparous Holstein (HOLS; n = 14) and Montbéliarde (MONT; n = 14) subjected to a sequential nutritional challenge of 4 successive 4-d 50% energy requirement periods (FR1, FR2, FR3, FR4) where FR = feed restriction. * Indicates that differences between d4 and d0 are significant ($P \leq 0.05$), or ● with tendency ($0.05 < P \leq 0.1$); When interaction FR × breed was significant ($P \leq 0.05$), HOLS and MONT are separately represented. HOLS + MONT (n = 28); HOLS (n = 14); MONT (n = 14).

compared with MONT across all the lactation stages, in accordance with the results obtained in the main experiment (Pires et al., 2025). Concentrations of the most abundant cytokines and chemokines were higher in HOLS. Although the effects of breed on inflammation modulation have been seldom studied, Lopreiato et al. (2019) reported higher concentrations of IL-1β and IL-6 in HOLS than in Simmental cows, a dual-purpose breed selected with lower energy intake and milk production than HOLS. Early lactating HOLS had lower plasma glucose and higher BHB than MONT, although BHB remained low in this study (Pires et al., 2025). Both energy metabolites can modulate immune activity (Horst et al., 2021). Furthermore, pro-inflammatory cytokines can be secreted by adipose tissue (Halberg et al., 2008; Häussler et al., 2022), and differences in fat depots and intensity of mobilization could explain breed differences in adaptative immune response. In the present study and subset of cows, breed differences in plasma NEFA at wk +2 did not reach significance. However, HOLS had lower BCS and greater plasma NEFA than MONT during early lactation when all cows and time points from the main experiment were analyzed (Pires et al., 2025), demonstrating breed differences in adipose tissue depots and mobilization.

Feed efficiency, assessed by RFI classification, does not appear to be associated with the regulation of cytokines and chemokines across lactation week, except for MIP-1β concentration, which was higher in less efficient animals (RFI+; i.e., cows with higher DMI than

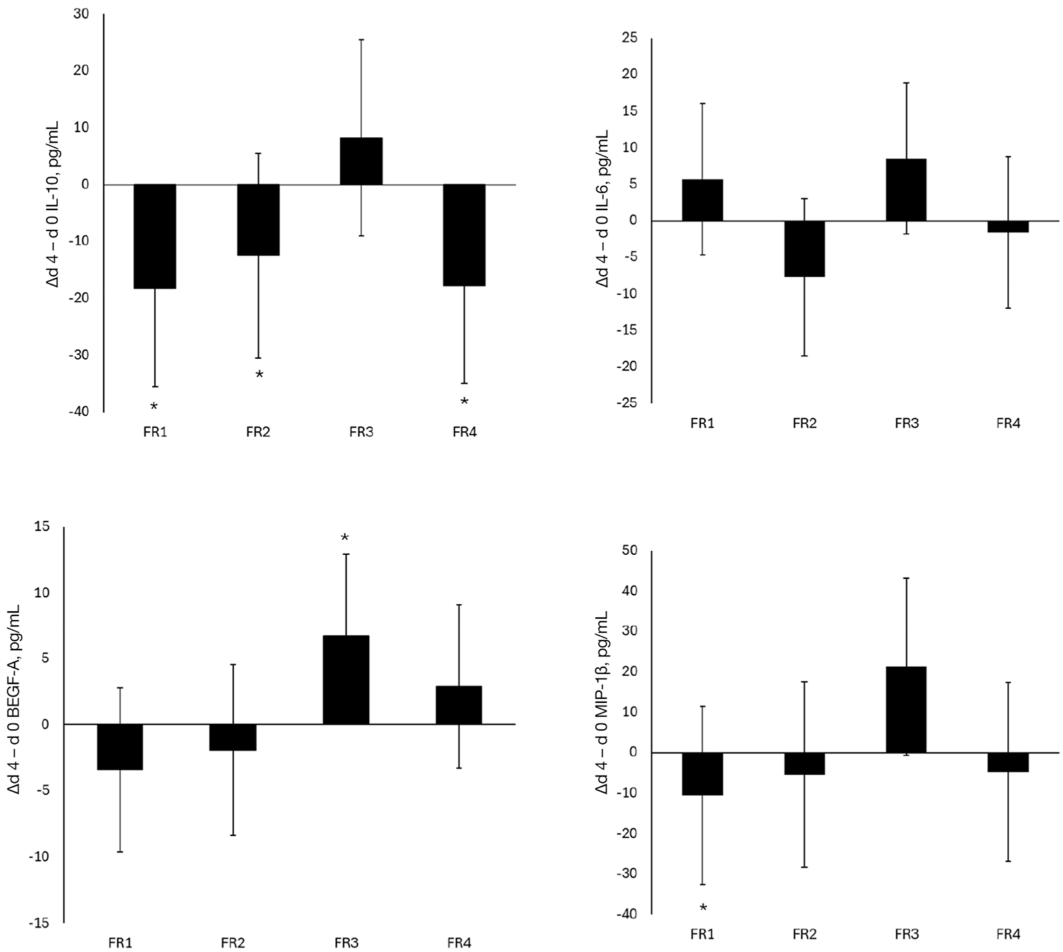

Figure 4 (Continued). LSM with 95% CI of cytokines and chemokines tumor necrosis factor α (TNFα), IL-1α, IL-10, vascular endothelial growth factor-A (VEGF-A), monocyte chemoattractant protein-1 or CCL2 (MCP-1), chemokine C-X-C pattern (IL-8), IL-6, and macrophage inflammatory protein −1β or CCL4 (MIP-1β), in multiparous Holstein (HOLS; n = 14) and Montbéliarde (MONT; n = 14) subjected to a sequential nutritional challenge of 4 successive 4-d 50% energy requirement periods (FR1, FR2, FR3, FR4) where FR = feed restriction. * Indicates that differences between d4 and d0 are significant ($P \leq 0.05$), or ● with tendency ($0.05 < P \leq 0.1$); When interaction FR × breed was significant ($P \leq 0.05$), HOLS and MONT are separately represented. HOLS + MONT (n = 28); HOLS (n = 14); MONT (n = 14).

expected). MIP-1β is mainly chemotactic for leukocytes and is involved in inflammation, but also plays roles in angiogenesis, hematopoiesis, and pathological conditions (Menten et al., 2002). Its expression has also been shown to increase with weight gain and obesity in humans (Surmi and Hasty, 2008). This could partly explain the slightly higher concentrations of MIP-1β in less feed efficient dairy cows, which may prioritize energy storage in body reserves over milk energy secretion. Despite the differences in MIP-1β concentrations between RFI+ and RFI−, no significant interaction between lactation week and RFI was observed. Therefore, and contrary to hypothesis, the regulation of innate IS from the periparturient period to mid-lactation does not appear to be largely influenced by breed or RFI classification under our experimental conditions.

Innate Immune Response to Successive FR Periods According to Breed and RFI

Despite systematic decreases in DMI, milk yield, and EB and increases in plasma NEFA, we observed significant but small changes in cytokine and chemokine concentrations that differed according to the FR period: FR1 decreased IL-1α, IL-10, MCP-1 and MIP-1β, whereas FR3 increased TNFα, IL-1α, VEGF-A, and IL-8. In contrast, Goetz et al. (2024) observed large increases in the concentrations of the pro-inflammatory cytokines IL-8, MCP-1, and VEGF-A in response to a single 3 d FR challenge at 40% of ER in mid-lactation Holstein cows. Lipomobilization, as indicated by plasma NEFA, increased similarly in the study of Goetz et al. (2024) (approximately 477 μ*M*, from 138 to 615 μ*M* after 3 d

Table 3. Cytokines and chemokines measured in plasma before (d 0) and after (d 4) 4-d feed restriction (FR) periods at 50% of energy requirement during the 4 sequential nutritional challenge (SNC) periods FR1, FR2, FR3, and FR4 in multiparous Holstein (HOLS; n = 14) and Montbéliarde (MONT; n = 14)[1]

	FR1		FR2		FR3		FR4		
Item[2]	d 0	d 4	d 0	d 4	d 0	d 4	d 0	d 4	d 7[3]
TNFα, pg/mL									
HOLS	4,637	4,291	3,646	3,673	2,523	2,766	2,512	2,536	2,256
Lower CI	2,491	2,305	2,127	2,154	1,505	1,649	1,487	1,502	1,321
Upper CI	8,634	7,990	5,165	5,192	4,230	4,637	4,242	4,283	3,852
MONT	2,514	2,589	2,655	2,503	1,733	1,861	1,617	1,650	1,368
Lower CI	1,350	1,390	1,216	1,334	1,034	1,110	958	977	801
Upper CI	4,681	4,820	4,255	4,373	2,907	3,120	2,732	2,786	2,336
FR effect[4]					B	A			
IL-1α, pg/mL									
HOLS	41.9[a]	38.6[b]	41.0	41.4	40.0	41.9	40.2	38.6	33.7
Lower CI	27.8	24.5	27.8	28.2	26.3	28.2	26.0	24.5	19.5
Upper CI	56.0	52.7	54.1	54.6	53.8	55.6	54.4	52.8	47.8
MONT	30.6[ab]	31.7[ab]	29.7	32.1	30.5	33.9	32.7	36.7	32.0
Lower CI	16.5	17.6	18.0	18.7	16.7	20.2	18.5	22.5	17.8
Upper CI	44.7	45.8	44.3	45.1	44.2	47.6	46.8	50.9	46.2
IL-10, pg/mL									
HOLS	294	262	144	129	143	164	139	114	115
Lower CI	204	173	106	91	93	115	102	84	85
Upper CI	383	352	182	167	192	213	187	154	156
MONT	205	200	125	115	116	111	100	88	89
Lower CI	116	111	87	77	67	62	74	65	66
Upper CI	295	290	163	153	165	160	135	119	120
FR effect[4]	A	B	A	B			A	B	B
VEGF-A, pg/mL									
HOLS	72.9	64.0	86.3	79.7	77.3	89.5	79.2	86.0	73.6
Lower CI	49.3	43.2	54.2	47.6	45.2	57.4	58.9	63.9	54.7
Upper CI	107.8	94.6	118.4	111.8	109.4	121.6	106.6	115.8	98.9
MONT	48.3	46.4	83.5	87.3	82.1	83.5	66.1	65.0	63.5
Lower CI	32.7	31.4	51.3	55.1	50.0	51.4	49.1	48.3	47.2
Upper CI	71.5	68.7	115.7	119.5	114.2	115.6	89.0	87.5	85.3
FR effect[4]					B	A			
MCP-1, pg/mL									
HOLS	801	727	656[a]	613[ab]	626	616	631	567	589
Lower CI	609	553	525	482	489	470	477	414	436
Upper CI	1,055	957	787	744	764	754	784	720	742
MONT	552	501	459[b]	485[ab]	500	512	499	451	476
Lower CI	420	380	328	353	363	375	346	298	323
Upper CI	727	659	591	616	637	650	652	604	630
FR effect[4]	A	B					A	B	A
IL-8, pg/mL									
HOLS	1,084*	851	942	631	473	760	518	860	595
Lower CI	762	598	631	320	328	527	336	557	393
Upper CI	1,542	1,211	1,253	943	683	1,098	799	1,326	903
MONT	554*	614	756	785	373	512	501	576	381
Lower CI	389	431	439	468	259	355	325	373	251
Upper CI	788	873	1,072	1,101	539	739	773	888	577
FR effect[4]					B	A	B	A	B
IL-6, pg/mL									
HOLS	21.1	28.6	6.80	8.39	19.4	38.3	10.5	6.0	4.5
Lower CI	−3.0	4.5	2.1	2.6	1.5	20.4	3.2	1.8	1.4
Upper CI	45.2	52.7	22.0	27.2	37.2	56.2	34.2	19.5	14.8
MONT	2.9	6.7	3.1	2.9	16.1	14.1	2.2	2.9	2.5
Lower CI	−21.1	−17.3	0.9	0.9	−1.8	−3.8	0.7	0.9	0.8
Upper CI	27.0	30.8	10.3	9.61	34.0	32.0	7.3	9.5	8.1
MIP-1β, pg/mL									
HOLS	88	77	108	109	94	106	94	87	79
Lower CI	68	60	76	77	71	80	72	67	60
Upper CI	114	100	141	141	124	140	122	113	103
MONT	97	92	104	93	91	96	89	86	81.7
Lower CI	75	71	72	60	69	72	68	66	62.5
Upper CI	125	119	136	125	121	126	115	112	106.9
FR effect[4]	A	B							

[a,b]Different superscripts (a–c) indicate significant ($P \leq 0.05$) FR × breed interaction.

[1]Data were analyzed using the MIXED procedure in SAS with the fixed effects of day (d 4 = feed restriction vs. d 0 = control), breed, and their interactions in 2 separate models. Cow was included as the random effect. Values are LSM with 95% CI. Log-transformed values (TNFα FR1, 3, and 4; IL-10 FR4; VEGF-A FR1 and 4; MCP-1 FR1; IL-8 FR1, 3, and 4; IL-6 FR2 and 4; MIP-1β FR1, 3, and 4) were back-transformed and presented as geometric means with 95% CI.

[2]MCP-1 = monocyte chemoattractant protein-1 or CCL2, MIP-1β = macrophage inflammatory protein −1β or CCL4, TNFα = tumor necrosis factor α, VEGF-A = vascular endothelial growth factor-A.

[3]d 7 = 3 d ad libitum after d 4.

[4]Different letters (A, B) indicate significant effect ($P \leq 0.05$) of feed restriction (FR) within an SNC FR period. The bottom row for each variable refers to statistical comparisons between d 0 and d 4 of FR across both breeds.

*For IL-8 in FR1, significant difference ($P \leq 0.05$) between HOLS and MONT.

at 40% ER) and in the present study (mean increase of 440 μ*M*, from 81 to 521 μ*M* after 4 d at 50% ER). Our study demonstrates that changes in metabolic status, characterized by increased NEFA and reduced EB, do not necessarily induce systemic inflammation. Consistent with this, clustering of early lactation cows based on EB temporal profiles showed limited associations between EB and IS, inflammation was instead associated with specific clinical events (Ma et al., 2024). Therefore, it is unlikely that negative EB and the associated increase in plasma NEFA, directly cause systemic inflammation. Previous studies suggested that FR could compromise the gut barrier integrity and cause systemic inflammation (Kvidera et al., 2017b; Horst et al., 2020), but given the small effects of the 4 FR periods on key cytokine and chemokine concentrations, we found no evidence of FR-induced inflammation in our model.

The Effects of Successive FR Periods

Cytokine and chemokine responses were not repeatable across successive FR periods. For instance, pro-inflammatory cytokine increases were observed only during FR3 (i.e., TNFα, VEGF-A, and IL-8), which might suggest a small cumulative effect of repeated FR. However, this effect was not observed during FR4, and cytokine concentrations at d 0 of each FR period remained consistently low, indicating no progressive increase in baseline levels from FR1 to FR4. Furthermore, the decrease in IL-10 after FR1, FR2, and FR4 supports the absence of a cumulative inflammatory response. This likely reflects a lack of activation of the anti-inflammatory pathway in the absence of a strong pro-inflammatory stimulus. The observed FR effects were also minor compared with those induced by intravenous or intramammary inflammatory challenges in mid-lactation dairy cows (Opgenorth et al., 2024a,b).

Breed or RFI had little effect on changes in innate IS during the FR: No systematic breed effects were observed in cytokine and chemokine responses during each FR, despite breed differences in nutrient prioritization toward milk secretion (Pires et al., 2025). Accordingly, feed efficiency had no significant effect on plasma cytokine and chemokine responses, except for MIP-1β, which decreased after FR1 only in the less efficient cows (i.e., RFI+). Contrary to our hypothesis, the successive FR did not indicate differences in inflammatory robustness among breeds or RFI groups.

CONCLUSIONS

Most cytokines decreased at wk +2 relative to prepartum and then showed similar profiles through wk +22, except IL-8, which increased at wk +2. Although breed differences were observed for TNFα, IL-10, MCP-1, and IL-8, with HOLS generally higher than MONT, innate IS followed similar temporal patterns across breeds during the periparturient period. Phenotypic feed efficiency did not modulate innate IS, except for higher MIP-1β in less efficient cows (RFI+). The SNC induced weak and nonreproducible changes in cytokine and chemokine concentrations, indicating that repeated short FR and the associated negative EB did not affect innate IS, despite consistent metabolic responses. Breed, phenotypic feed efficiency, and repeated feed restrictions did not strongly modulate innate IS.

NOTES

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Nonstandard abbreviations used: EB = energy balance; ER = energy requirements; FR = feed restriction; FRAP = ferric reducing ability of plasma; HOLS = Holstein; IS = inflammatory status; LLOQ = low limit of quantification; LOD = limit of detection; MONT = Montbéliarde; NEFA = nonesterified fatty acids; RFI = residual feed intake; SNC = sequential nutritional challenge.

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